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Portable Air Conditioner with Decontamination Plant

Abstract

The present application discloses a portable air conditioner with a decontamination plant, including an air conditioner body, a decontamination component and a clamping component. A main frame in the decontamination component is embedded and mounted in a slotted hole provided on one side of the air conditioner body, and a filter layer and a meltblown layer are embedded and mounted in a through slot provided on the main frame. The present application enables air conditioners to filter and remove dust and bacteria in the air by means of a decontamination component so as to enable portable air conditioners to improve the air quality while adjusting the temperature. The mounting of the decontamination component is carried out by means of a clamping component, which is easy to dismantle and mount, so as to improve the maintenance efficiency of the decontamination plant in the later stage.

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Background/Summary

TECHNICAL FIELD

[0001] The present application relates to the technical field of air conditioners, and more particularly, to a portable air conditioner with a decontamination plant.

BACKGROUND ART

[0002] A portable air conditioner, also known as a mobile air conditioner, is a refrigerating device which can be freely moved and placed; and the main features of portable air conditioners are small size, light weight, easy to move to any place requiring refrigeration, and requiring no fixed mounting. Compared with traditional air conditioners, portable air conditioners have lower energy consumption, which helps to save energy and reduce emissions. The existing portable air conditioners basically meet the needs of users, but there are still certain shortcomings. When used, the existing portable air conditioners can only decontaminate the air relatively roughly, and even have no air decontamination function. At the same time, the decontamination plants are often fixed by screws, which is time-consuming and laborious to assemble, and inconvenient to maintain in the later stage. Therefore, it is necessary to design a portable air conditioner with a decontamination plant.

SUMMARY OF THE INVENTION

[0003] It is an object of the present application to provide a portable air conditioner with a decontamination plant for solving the disadvantages that the existing portable air conditioners can only decontaminate the air relatively roughly, and are unable to remove bacteria from the environment; and that meantime, the decontamination plants are time-consuming and laborious to mount, and are inconvenient to maintain in the later stage.

[0004] In order to solve the above-mentioned technical problem, the present application provides the following technical solutions: a portable air conditioner with a decontamination plant, including an air conditioner body, a decontamination component and a clamping component, wherein a main frame in the decontamination component is embedded and mounted in a slotted hole provided on one side of the air conditioner body, a filter layer and a meltblown layer are embedded and mounted in a through slot provided on the main frame, a clamping block in the clamping component is clamped in an inclined slot provided on one side of the main frame, the clamping block is slidably connected in the slotted hole provided on one side of the air conditioner body, a rotary rod is rotatably connected in the slotted hole provided on one side of the clamping block, a fixing rod is fixedly connected to the rotary rod, and a handle is fixedly connected to the fixing rod.

[0005] According to a further technical solution of the present application, an air outlet is provided on a front face of the air conditioner body.

[0006] According to a further technical solution of the present application, the bottom of the air conditioner body is provided with a support foot.

[0007] According to a further technical solution of the present application, the decontamination component is composed of a main frame, an air inlet, a filter layer, a meltblown layer, a ventilation slot and an ultraviolet lamp, wherein the main frame is provided with an air inlet, the main frame is provided with a ventilation slot therein, and the ultraviolet lamp is provided on one side of the ventilation slot.

[0008] According to a further technical solution of the present application, the clamping component is composed of a clamping block, a rotary rod, a fixing rod, a handle, a connecting plate, a spring member, a backing plate and a support member, wherein a support member is slidably connected in a through slot provided on one side of the clamping block, the support member is fixedly connected in a slotted hole provided on one side of the air-conditioning main body, a connecting plate is fixedly connected to one side of the clamping block, the spring member is fixedly connected to the connecting plate, one end of the spring member is fixedly connected to the backing plate, and the backing plate is fixedly connected in the slotted hole provided on the air conditioner body.

[0009] According to a further technical solution of the present application, a housing is provided outside the air conditioner body, the housing has a cavity, a first air inlet and a first air outlet are provided on an outer wall of the housing to communicate the cavity with the outside, a first air duct for air communication is arranged between the first air inlet and the first air outlet, a first grid is provided at the first air inlet, a semiconductor chilling plate is mounted inside the housing, and one face of the semiconductor chilling plate is in communication with the first air duct for adjusting the air temperature in the first air duct.

[0010] According to a further technical solution of the present application, a first fan is mounted inside the first air duct, and the air supply direction faces the first air outlet; a magnet is mounted at the first air outlet; a power supply is mounted inside the housing, and is respectively connected to the semiconductor chilling plate and the first fan via a conductive member; several separator plates are arranged inside the housing to divide the cavity into a battery compartment, a first air duct and a cooling compartment, and the power supply is mounted in the battery compartment; and the first air duct is adjacent to the cooling compartment.

[0011] According to a further technical solution of the present application, a first cooling plate is disposed in the first air duct, and the first cooling plate is in contact with a face of the semiconductor chilling plate that communicates with the first air duct.

[0012] According to a further technical solution of the present application, a second cooling plate is arranged in the cooling compartment, and the second cooling plate is in contact with a face of the semiconductor chilling plate facing the inside of the cooling compartment, and a second fan is arranged in the cooling compartment and is connected to a power supply via a conductive member; a second air inlet is also provided on the compartment wall of the cooling compartment, the air supply direction of the second fan faces the cooling vent, and a second grid is provided at the second air inlet.

[0013] According to a further technical solution of the present application, a second air duct is provided on one side of the housing, the second air duct is of a cylindrical shape with an opening, one end of the second air duct is provided with a second air outlet, the second air outlet is greater than the cooling vent, and the second air outlet covers the cooling vent and is detachably connected to the housing; and a filter screen is arranged in the first air duct, and a socket is provided on the outer wall of the housing.

[0014] The present application provides a portable air conditioner with a decontamination plant, which is advantageous in that: the dust in the air is filtered through the filter layer, the bacteria in the air is removed by the meltblown layer, and then the remaining bacteria are removed by using the ultraviolet lamp, so that the air conditioner can filter and remove the dust and the bacteria in the air. Therefore, the portable air conditioner improves the air quality while adjusting the temperature. When the decontamination component is mounted, the decontamination component is aligned and then pushed in, so that two sides thereof slide on the inclined plane provided by the clamping block. The clamping block is retracted, and the spring member is compressed via the connecting plate. After it is pushed in place, the spring force of the spring member is released, so that the clamping block is clamped in the inclined slot provided on the side face of the main frame, thereby completing the fixation of the decontamination component. The mounting is quick and convenient, without fixing via a screw. When it is necessary to remove the decontamination component, the handle is pulled to drive the clamping block to retract, and at the same time, the spring member is compressed, and then rotation is performed to drive the rotary rod to rotate. After the fixing rod is aligned with a limiting slot provided on the support member, the handle is loosened, so that the fixing rod is clamped on the support member. After the operation of the handle on both sides is completed, the decontamination component can be taken out, and the removal is simple, so as to improve the maintenance efficiency of the decontamination plant at a later stage. The semiconductor chilling plate is used by air conditioners as a temperature adjustment mechanism to replace the micro-compressor in the prior art, so that the power consumption is reduced, and then the built-in battery with a low power storage can be used. Compared with the micro-compressor and the mobile power supply, the use of the semiconductor chilling plate and the built-in battery reduces the overall weight.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] In order to illustrate the embodiments of the present application or the technical solutions in the prior art more clearly, the following will briefly introduce the drawings that need to be used in the description of the embodiments or the prior art. Obviously, the drawings in the following description are some embodiments of the present application. For those of ordinary skills in the art, other drawings can be obtained according to these drawings without involving inventive efforts.

[0016] FIG. **1** is a three-dimensional view showing an overall structure of the present application;

[0017] FIG. **2** is a schematic view showing a position and a structure of an air inlet according to the present application;

[0018] FIG. **3** is a schematic view showing a structure of a decontamination component according to the present application;

[0019] FIG. **4** is a schematic view showing a structure of region A in FIG. **3**;

[0020] FIG. **5** is a schematic view showing a position and a structure of an ultraviolet lamp according to the present application;

[0021] FIG. **6** is a schematic view showing an internal structure of an air conditioner body according to the present application;

[0022] FIG. **7** is a schematic view showing a position and a structure of a socket according to the present application; and

[0023] FIG. **8** is a schematic view showing a structure of a second air duct according to the present application.

[0024] In the figures: **1**, air conditioner body; **2**, decontamination component; **3**, clamping component; **4**, third air outlet; **5**, support foot; **21**, main frame; **22**, third air inlet; **23**, filter layer; **24**, meltblown layer; **25**, ventilation slot; **26**, ultraviolet lamp; **31**, clamping block; **32**, rotary rod; **33**, fixing rod; **34**, handle; **35**, connecting plate; **36**, spring member; **37**, backing plate; **38**, support member; **100**, housing; **110**, first air inlet; **111**, first grid; **120**, first air outlet; **130**, separator plate; **200**, semiconductor chilling plate; **300**, first air duct; **310**, first fan; **320**, first cooling plate; **330**, filter screen; **400**, power supply; **500**, cooling compartment; **510**, second cooling plate; **520**, second fan; **530**, cooling vent; **540**, second air inlet; **541**, second grid; **600**, second air duct; **610**, second air outlet; **700**, socket; **800**, magnet.

DETAILED DESCRIPTION OF THE INVENTION

[0025] In order to make the purposes, technical solutions, and advantages of the embodiments of the present application clearer, the technical solutions in the embodiments of the present application will be clearly and completely described below in conjunction with the accompanying drawings in the embodiments of the present application. Obviously, the described embodiments are a part of the embodiments of the present application, rather than all the embodiments. Based on the embodiments of the present application, all other embodiments obtained by one of ordinary skills in the art without involving any inventive effort are within the scope of the present application.

[0026] With reference to FIGS. **1-8**, an embodiment of the present application is provided: a portable air conditioner with a decontamination plant, including an air conditioner body **1**, a decontamination component **2** and a clamping component **3**, wherein a main frame **21** in the decontamination component **2** is embedded and mounted in a slotted hole provided on one side of the air conditioner body **1**, a filter layer **23** and a meltblown layer **24** are embedded and mounted in a through slot provided on the main frame **21**, a clamping block **31** in the clamping component **3** is clamped in an inclined slot provided on one side of the main frame **21**, the clamping block **31** is slidably connected in the slotted hole provided on one side of the air conditioner body **1**, a rotary rod **32** is rotatably connected in the slotted hole provided on one side of the clamping block **31**, a fixing rod **33** is fixedly connected to the rotary rod **32**, a handle **34** is fixedly connected to the fixing rod **33**, and a third air outlet **4** is provided on a front face of the air conditioner body **1**.

[0027] A support foot **5** is provided at the bottom of the air conditioner body **1**; the decontamination component **2** is composed of a main frame **21**, a third air inlet **22**, a filter layer **23**, a meltblown layer **24**, a ventilation slot **25** and an ultraviolet lamp **26**; the third air inlet **22** is provided on the main frame **21**; the ventilation slot **25** is provided inside the main frame **21**; and the ultraviolet lamp **26** is provided on one side of the ventilation slot **25**. The clamping component **3** is composed of a clamping block **31**, a rotary rod **32**, a fixing rod **33**, a handle **34**, a connecting plate **35**, a spring member **36**, a backing plate **37** and a support member **38**. The support member **38** is slidably connected in a through slot provided on one side of the clamping block **31**, the support member **38** is fixedly connected in a slotted hole provided on one side of the air conditioner body **1**, the connecting plate **35** is fixedly connected to one side of the clamping block **31**, a spring member **36** is fixedly connected to the connecting plate **35**, one end of the spring member **36** is fixedly connected to the backing plate **37**, the backing plate **37** is fixedly connected in the slotted hole provided on the air conditioner body **1**, a housing **100** is provided outside the air conditioner body **1**, the housing **100** has a cavity, and a first air inlet **110** and a first air outlet **120** are provided on an outer wall of the housing **100** for communicating the cavity with the outside. A first air duct **300** for air circulation is arranged between the first air inlet **110** and the first air outlet **120**, a first grid **111** is arranged at the first air inlet **110**, a semiconductor chilling plate **200** is fitted inside the housing **100**, and one face of the semiconductor chilling plate **200** is in communication with the first air duct **300** for adjusting the air temperature in the first air duct **300**.

[0028] A first fan **310** is mounted inside the first air duct **300**, its air supply direction faces the first air outlet **120**, a magnet **800** is fitted at the first air outlet **120**, a power supply **400** is fitted inside the housing **100** and is respectively connected to the semiconductor chilling plate **200** and the first fan **310** via a conductive member, several separator plates **130** are arranged inside the housing **100** to divide the cavity into a battery compartment, a first air duct **300** and a cooling compartment **500**, the power supply **400** is fitted in the battery compartment, and the first air duct **300** is adjacent to the cooling compartment **500**. A first cooling plate **320** is arranged in the first air duct **300**, and the first cooling plate **320** is in contact with one face of the semiconductor chilling plate **200** which is in communication with the first air duct **300**. A second cooling plate **510** is arranged in the cooling compartment **500**, the second cooling plate **510** is in contact with one face of the semiconductor chilling plate **200** which faces the inside of the cooling compartment **500**, and a second fan **520** is arranged in the cooling compartment **500** and is connected to the power supply **400** via a conductive member. The compartment wall of the cooling compartment **500** is further provided with a second air inlet **540**, the air supply direction of the second fan **520** faces the cooling vent **530**, a second grid **541** is provided at the second air inlet **540**, a second air duct **600** is provided on one side of the housing **100**, the second air duct **600** is of a cylindrical shape with an opening, one end of the second air duct **600** is provided with a second air outlet **610**, the second air outlet **610** is greater than the cooling vent **530**, and the second air outlet **610** covers the cooling vent **530** and is detachably connected to the housing **100**. A filter screen **330** is arranged in the first air duct **300**, and a socket **700** is provided on an outer wall of the housing **100**.

[0029] Specifically, in use, first, the dust in the air is filtered through the filter layer **23**, the bacteria in the air is removed by the meltblown layer **24**, and then the remaining bacteria are removed by using the ultraviolet lamp **26**, so that the air conditioner can filter and remove the dust and the bacteria in the air. Therefore, the portable air conditioner improves the air quality while adjusting the temperature.

[0030] When the decontamination component **2** is mounted, the decontamination component **2** is aligned and then pushed in, so that two sides thereof slide on the inclined plane provided by the clamping block **31**. The clamping block **31** is retracted, and the spring member **36** is compressed via the connecting plate **35**. After it is pushed in place, the spring force of the spring member **36** is released, so that the clamping block **31** is clamped in the inclined slot provided on the side face of the main frame **21**, thereby completing the fixation of the decontamination component **2**. The mounting is quick and convenient, without fixing via a screw. When it is necessary to remove the decontamination component **2**, the handle **34** is pulled to drive the clamping block **31** to retract, and at the same time, the spring member **36** is compressed, and then rotation is performed to drive the rotary rod **32** to rotate. After the fixing rod **33** is aligned with a limiting slot provided on the support member **38**, the handle **34** is loosened, so that the fixing rod **33** is clamped on the support member **38**. After the operation of the handle **34** on both sides is completed, the decontamination component **2** can be taken out, and the removal is simple, so as to improve the maintenance efficiency of the decontamination plant at a later stage.

[0031] The battery **400** is started, and the battery **400** supplies power to the semiconductor chilling plate **200** and the first fan **310**, respectively, so that the temperature of the cold side of the semiconductor chilling plate **200** decreases while the temperature of the hot side increases. When the cold side of the semiconductor chilling plate **200** is in communication with the first air duct **300**, the air in the first air duct **300** decreases with the decrease of the temperature of the cold side, forming a low-temperature air, and, under the action of the first fan **310**, forming a low-temperature air flowing out of the first air outlet **120**, thereby achieving the effect of external refrigeration. When the heat plate of the semiconductor chilling plate **200** is in communication with the first air duct **300**, the air in the first air duct **300** increases with the increase of the temperature of the hot side, forming a high-temperature air, and under the action of the first fan **310**, forming a high-temperature air flowing out of the first air outlet **120**, so as to achieve the effect of external heating. The semiconductor chilling plate **200** is used as a temperature adjustment mechanism to replace the micro-compressor in the prior art, so that the power consumption is reduced, and then the built-in battery **400** with a low power storage can be used. Compared with the micro-compressor and the mobile power supply, the use of the semiconductor chilling plate **200** and the built-in battery **400** reduces the overall weight and solves the technical problem that the mobile air-conditioning device is heavy in the prior art.

[0032] In describing the present application, it needs to be noted that unless otherwise expressly specified and limited, the terms “mounted” “connected”, and “connecting” should be interpreted broadly. For example, it can be a fixed connection or a detachable connection, or an integrated connection; it can be a mechanical connection, or an electrical connection; it can be directly connected or indirectly connected through an intermediate medium, and it can be the communication within two elements. For a person of ordinary skills in the art, the specific meaning of the above-mentioned terms according to the present application can be understood according to specific situations.

[0033] The above-described plant embodiments are merely illustrative. The units described as separate components may or may not be physically separated, and the components displayed as units may or may not be physical units, that is, they may be located in one place, or they may be distributed on multiple network units. Some or all of the modules can be selected according to actual needs to achieve the purpose of the scheme of the embodiment. A person of ordinary skills in the art would understand and practice that without involving any inventive effort.

[0034] Finally, it should be noted that: the above embodiments are provided only to illustrate the technical solutions of the present application, not to limit it; while the present application has been described in detail with reference to the foregoing embodiments, those of ordinary skills in the art should understand that: the technical solutions disclosed in the above-mentioned embodiments can still be modified, or some of the technical features can be replaced by equivalents; such modifications and substitutions do not make the essence of the corresponding technical solutions depart from the spirit and scope of the technical solutions of the embodiments of the present application.

Claims

1. A portable air conditioner with a decontamination plant, comprising an air conditioner body (**1**), a decontamination component (**2**) and a clamping component (**3**), wherein a main frame (**21**) of the decontamination component (**2**) is embedded and mounted in a slotted hole provided on one side of the air conditioner body (**1**), a filter layer (**23**) and a meltblown layer (**24**) are embedded and mounted in a through slot provided on the main frame (**21**), a clamping block (**31**) of a clamping component (**3**) is clamped in an inclined slot provided on one side of the main frame (**21**), the clamping block (**31**) is slidably connected in the slotted hole provided on one side of the air conditioner body (**1**), a rotary rod (**32**) is rotatably connected in the slotted hole provided on one side of the clamping block (**31**), a fixing rod (**33**) is fixedly connected to the rotary rod (**32**), and a handle (**34**) is fixedly connected to the fixing rod (**33**).
2. The portable air conditioner with a decontamination plant according to claim 1, wherein a third air outlet (**4**) is provided on a front face of the air conditioner body (**1**).
3. The portable air conditioner with a decontamination plant according to claim 2, wherein a support foot (**5**) is provided at a bottom of the air conditioner body (**1**).
4. The portable air conditioner with a decontamination plant according to claim 1, wherein the decontamination component (**2**) is composed of a main frame (**21**), a third air inlet (**22**), a filter layer (**23**), a meltblown layer (**24**), a ventilation slot (**25**) and an ultraviolet lamp (**26**), and wherein the main frame (**21**) is provided with the third air inlet (**22**), an inside of the main frame (**21**) is provided with the ventilation slot (**25**), and the ultraviolet lamp (**26**) is provided on one side of the ventilation slot (**25**).
5. The portable air conditioner with a decontamination plant according to claim 1, wherein the clamping component (**3**) is composed of a clamping block (**31**), a rotary rod (**32**), a fixing rod (**33**), a handle (**34**), a connecting plate (**35**), a spring member (**36**), a backing plate (**37**) and a support member (**38**), and wherein the support member (**38**) is slidably connected in a through slot provided on one side of the clamping block (**31**), the support member (**38**) is fixedly connected in a slotted hole provided on one side of the air conditioner body (**1**), the connecting plate (**35**) is fixedly connected on one side of the clamping block (**31**), the spring member (**36**) is fixedly connected on the connecting plate (**35**), one end of the spring member (**36**) is fixedly connected to the backing plate (**37**), and the backing plate (**37**) is fixedly connected to the slotted hole provided in the air conditioner body (**1**).
6. The portable air conditioner with a decontamination plant according to claim 5, wherein a housing (**100**) is provided on an outside of the air conditioner body (**1**), the housing (**100**) has a cavity, a first air inlet (**110**) and a first air outlet (**120**) are provided on an outer wall of the housing (**100**) to communicate the cavity with the outside, a first air duct (**300**) is arranged between the first air inlet (**110**) and the first air outlet (**120**) for air circulation, a first grid (**111**) is provided at the first air inlet (**110**), a semiconductor chilling plate (**200**) is mounted inside the housing (**100**), and one face of the semiconductor chilling plate (**200**) is in communication with the first air duct (**300**) for adjusting air temperature in the first air duct (**300**).
7. The portable air conditioner with a decontamination plant according to claim 6, wherein the first air duct (**300**) is internally fitted with a first fan (**310**), with an air supply direction thereof facing a first air outlet (**120**), a magnet (**800**) is fitted at the first air outlet (**120**), the housing (**100**) is internally fitted with a power supply (**400**), the power supply (**400**) is respectively connected to the semiconductor chilling plate (**200**) and the first fan (**310**) via a conductive member, several

separator plates (**130**) are arranged in the housing (**100**), the cavity is divided into a battery compartment, a first air duct (**300**) and a cooling compartment (**500**), the power supply (**400**) is fitted in the battery compartment, and the first air duct (**300**) is adjacent to the cooling compartment (**500**).

8. The portable air conditioner with a decontamination plant according to claim 7, wherein a first cooling plate (**320**) is arranged in the first air duct (**300**), and the first cooling plate (**320**) is in contact with a face of the semiconductor cooling plate (**200**) which communicates with the first air duct (**300**).

9. The portable air conditioner with a decontamination plant according to claim 7, wherein a second cooling plate (**510**) is arranged in the cooling compartment (**500**), the second cooling plate (**510**) is in contact with a face of the semiconductor chilling plate (**200**) facing an inside of the cooling compartment (**500**), and a second fan (**520**) is arranged in the cooling compartment (**500**) and is connected to the power supply (**400**) via a conductive member; and a second air inlet (**540**) is further provided on a compartment wall of the cooling compartment (**500**), an air supply direction of the second fan (**520**) faces the cooling vent (**530**), and a second grid (**541**) is provided at the second air inlet (**540**).

10. The portable air conditioner with a decontamination plant according to claim 7, wherein a second air duct (**600**) is provided on one side of the housing (**100**), the second air duct (**600**) is of a cylindrical shape with an opening, one end of the second air duct (**600**) is provided with a second air outlet (**610**), the second air outlet (**610**) is larger than the cooling vent (**530**), the second air outlet (**610**) covers the cooling vent (**530**) and is detachably connected to the housing (**100**), a filter screen (**330**) is arranged in the first air duct (**300**), and a socket (**700**) is provided on an outer wall of the housing (**100**).